

Education

STI College Caloocan

Bachelor of Science in Computer Science

Expected Graduation: July, 2026

Eulogio Rodriguez Vocational High School

Computer System Servicing

2022

Experience

PopCulture Community Eastwood

Quezon City

Full-Stack Developer Intern (300-Hr OJT)

February, 2026 – May, 2026

- Developed backend systems for a multi-platform renter management and e-commerce app supporting up to **60 renters**
- Designed **Supabase RPCs**, **Edge Functions**, and transactional workflows across web and **Flutter** mobile applications
- Built **ledger-based** finance and inventory tracking using delta-driven updates for transactional consistency
- Integrated cloud infrastructure and services including **Google Cloud**, **Firebase**, **Cloud Run**, and automated email reporting for renter earnings
- Coordinated directly with clients to translate business requirements into technical implementations

Leadership & Activities

CREOTECH

IT TEAM LEADER – IT DEPARTMENT (80-Hr Work Immersion)

2022

- Led a **38-member IT department**, facilitating brainstorming sessions and coordinating task delegation for app development projects
- Developed an employee attendance tracker and payroll calculator using MIT App Inventor
- Compiled memorandums and individual task reports submitted to supervisors, and monitored team attendance and compliance

Projects

A.EYE - AI-POWERED ASSISTIVE IOT SYSTEM (Thesis)

Team Leader/Lead Developer

- Built a **Raspberry Pi-based assistive device** using a chest-mounted camera and **YOLOv8s** model to detect and teach braille letters in real time
- Compiled and labeled a **5,000-image braille dataset** using **Roboflow** and **CVAT**, including manually embossed paper samples for training data
- Implemented **ArUco** marker-based spatial anchoring to stabilize detection output against camera movement and viewing angle.
- Designed two interaction modes — a guided teach mode and an interactive quiz mode — with correct responses logged to a database for teacher and parent tracking
- Achieved **95% detection accuracy** on controlled test surfaces; identified thermal throttling and low-contrast embossing as key hardware constraints for future iteration

Technical Skills

Programming Languages: Python, Java, C#, Dart

Frameworks & Libraries: Flutter, OpenCV, Ultralytics

Database & Backend: Firebase, Supabase, PostgreSQL

AI & ML: Computer Vision, AI Model Training & Deployment, Roboflow, CVAT

Tools & Platforms: Git & GitHub, Android Studio, Linux/Terminal, Raspberry Pi & IOT Systems